Critic and User Reviews on Video Game Commercial Success

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1. Abstract

This project investigates the relationship between video game review scores and commercial success using a dataset containing global sales, critic scores, critic count, user scores, user count and game metadata such as genre and platform. Exploratory data analysis was conducted using visualizations including scatter plots, boxplots, and correlation heatmaps to identify patterns between ratings and sales. The results of this analysis suggest a positive but weak relationship between review scores and global sales, with significant variability influenced by factors such as platform and genre. An interactive Plotly dashboard was developed to visualize the distribution of global sales across platforms, allowing users to explore the data through filtering and highlighting key trends. To further analyze predictive relationships, multiple machine learning models including Linear Regression, Decision Tree, and Random Forest were trained and evaluated using N-fold cross-validation. The models were assessed using RMSE and the best-performing model was tested on a held-out test set. Overall, the findings indicate that while review scores contribute to predicting sales, additional factors are necessary to fully explain commercial success in the video game industry.

2. Introduction

2.1 Inspiration and Problem Definition

Despite the video game industry generating over $180 billion annually, with some studios investing millions of dollars into the development and marketing of their games, some game's commercial success still crashes and burns. Some recent examples include Onrush, Anthem, Suicide Squad, and the biggest offender, Concord. Many would argue that a fairly large reason these games failed as hard as they did were because of public perception which were shaped by critic and user scores and reviews on the games. However the actual correlation between scores and a game's financial performance is still unclear, which is the inspiration for this project.

The central problem we investigated was:

Can review scores and related data be used to predict a game's global sales performance?

To help with our research problem, we looked at the following five questions:

1. Is there a significant correlation between critic scores and global sales?
2. Are user scores more predictive of commercial success than critic scores?
3. Does the relationship between ratings and sales vary by genre?
4. Has the predictive power of review scores changed over the years?
5. How does the distribution of global video game sales differ across platforms?

We also tested to see how accurate the machine learning models we trained using the variables of the database were in determining the global sales of a game.

2.2 Background, Ideas, Goals

Gaming is the largest entertainment industry in the world, pulling in more revenue than films and music combined. What is fascinating about this industry is how impactful the public perspective can be on a games lifespan. This makes it difficult to understand what determines a

‘good’ game. Filtering out what variables truly drive sales is imperative to succeeding in this competitive industry. An interesting trend as of late has been the correlation between public relation disasters and a games success. When a game fails it tends to fail hard, Concord a game with rumored budgets in the hundreds of millions lasted merely a few weeks before permanently going offline becoming one of the largest failures in modern gaming history.

Our goal was to identify if any correlation exists between a games critic and user scores and its actual sales performance. To also identify which of the two scores is most indicative of performance as if there is any way to predict a game's sales based on its reviews.

3. Initial Data Exploration

3.1 Dataset Preparation

The dataset used for this project was obtained from Kaggle and contains information on video game sales, such as regional and global sales, sourced from VGChartz, and review scores and metadata such as genre and platform that were scraped from Metacritic. It contains more than ten thousand unique games, with about seven thousand of those having no missing values. It was selected for its size and relevant columns such as critic and user scores, and genre and platform.

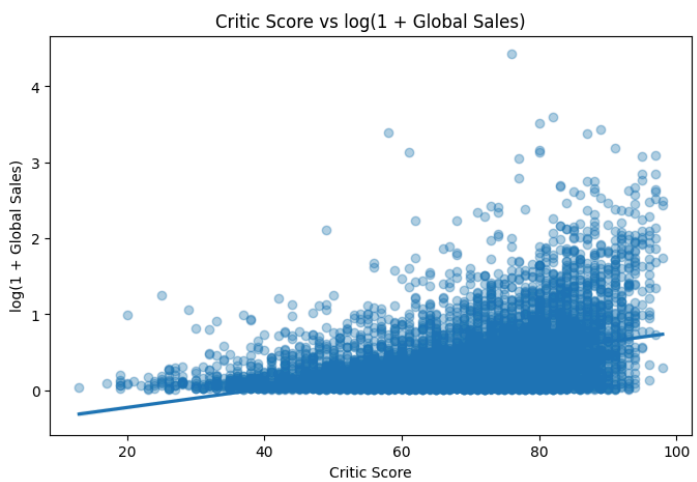
The dataset was first inspected for missing values and data types. Columns that were not relevant to the analysis, such as Name, Publisher, Developer, and Rating, were identified and excluded from certain parts of the analysis due to their high cardinality or limited usefulness. Our key variables were:

- **Global Sales:** Total worldwide sales (in millions)
- **Critic Score:** The score of the game from critics
- **Critic Count:** The number of critical reviews for the game

- **User Score:** The score of the game from users
- **User Count:** The number of user reviews for the game
- **Year of Release:** The release year of the game
- **Genre:** What type or genre of game it is
- **Platform:** What console the game was released on

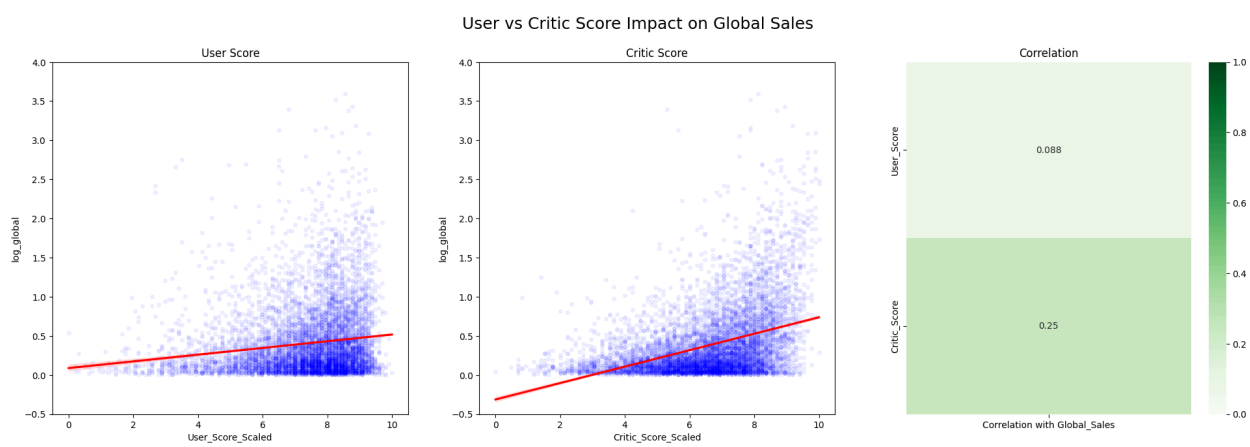
3.2 Research Problem Plots

1. Is there a significant correlation between critic scores and global sales?



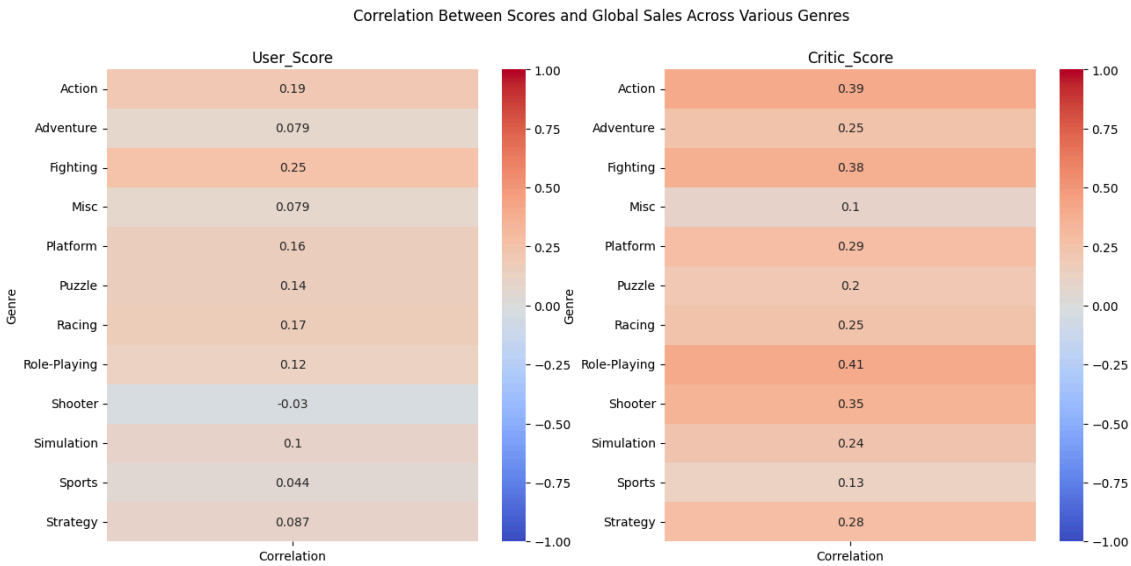
Result: This plot shows that there is a moderate correlation of higher critic scores being associated with slightly higher sales on average.

2. Are user scores more predictive of commercial success than critic scores?



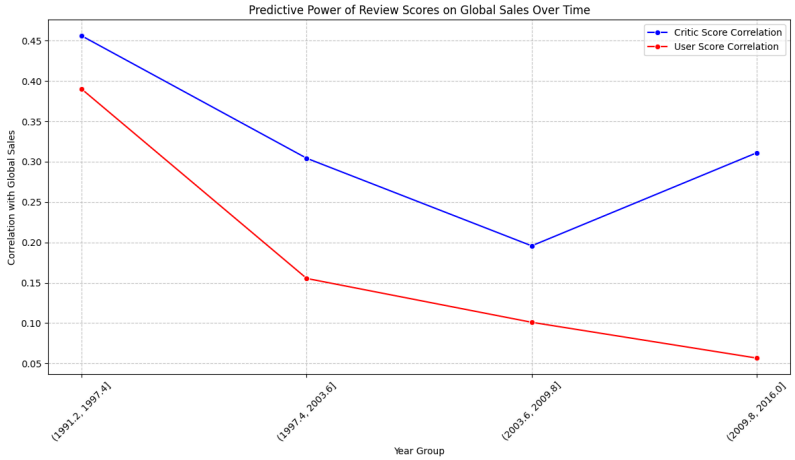
Result: The association between Critic_Score and Global_Sales is stronger than the association between User_Score and Global_Sales.

3. Does the relationship between ratings and sales vary by genre?



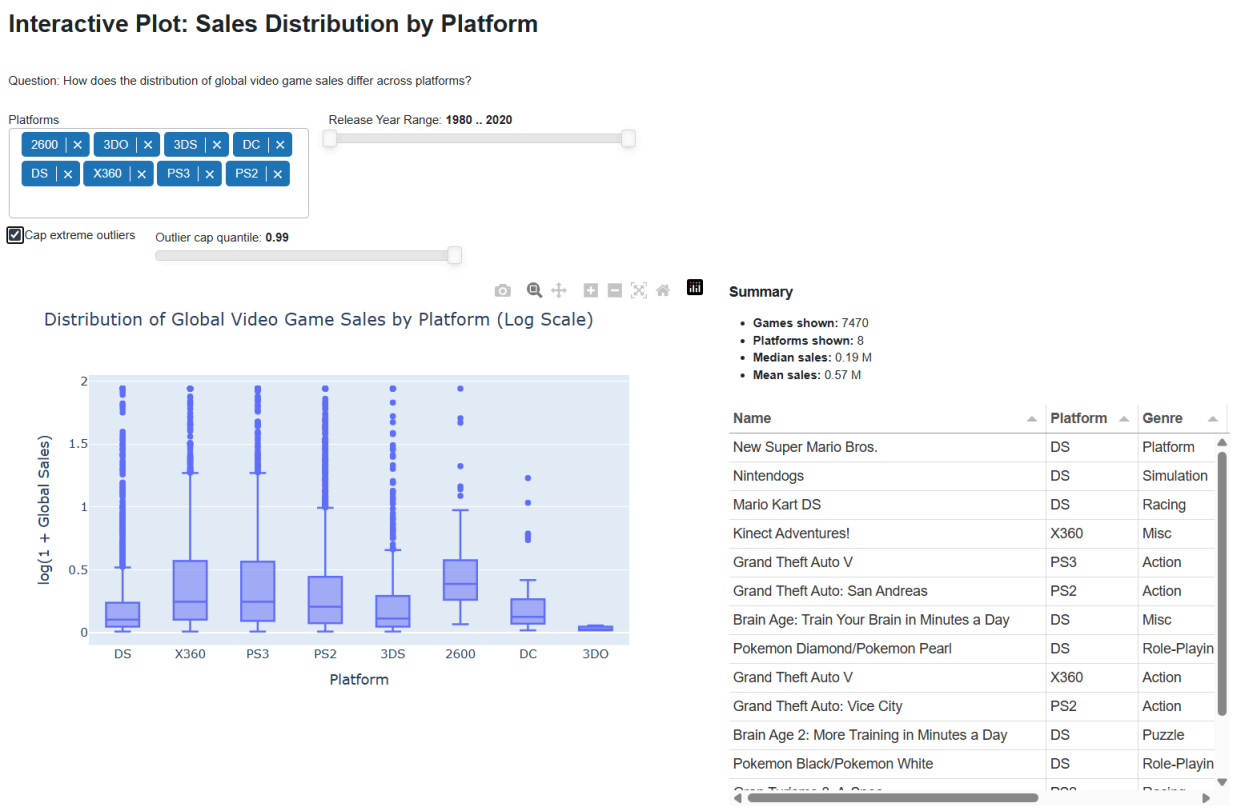
Result: Fighting and action games generally had higher correlation between scores and true global sales. For critics, the correlation between misc games and sports games were much lower. Similarly, for user scores shooters and sports games had less correlation, although user scores had less correlation with sales in general.

4. Has the predictive power of review scores changed over the years?



Result: The predictive power of review scores were on a downward trend for the past few decades, however recently there has been a rise in the critic score correlation.

5. How does the distribution of global video game sales differ across platforms?



Result: Home consoles such as the Xbox 360, PS3, and PS2, tend to have higher median and variable global sales than handheld consoles and PCs, which show lower average sales and less high selling titles.

Our interactive plot has options that filter data for the interactive plot and table. This includes a toggle to cap extreme outliers, a multi-choice panel for selecting multiple platforms, a year slider, a table for showing some of the games in the filtered data, and a summary that contains information like the number of games shown and sales.

The exploratory visualizations revealed that the relationship between review scores and global sales is weak and varies across features such as genre and platform. Additionally, the

distribution of global sales is highly skewed, and the relationships between variables are not strictly linear. Based on these observations, we selected a combination of models, including Linear Regression as a baseline and more flexible non-linear models such as Decision Tree and Random Forest, to better capture complex patterns in the data.

4. Machine Learning

4.1 Data Preparation & Pipeline

The dataset was first cleaned by handling missing values and selecting relevant features for analysis. Columns such as Name, Publisher, Developer, and Rating were excluded due to high cardinality and limited predictive value. Regional sales columns were also removed to avoid data leakage with Global_Sales.

Feature engineering was done to improve model performance. An average score (**Avg_Score**) was calculated by combining critic and user scores, and a total review count (**Total_Count**) was created by summing critic and user counts. These features were created to better capture overall game reception and popularity.

The dataset was split into train and test sets using an 80/20 split for an unbiased evaluation. The training set was used for model fitting and cross validation, while the test set was reserved for the final evaluation.

We implemented a preprocessing pipeline using Scikit-learn's ColumnTransformer to handle both numerical and categorical data. We used median imputation to handle missing values in the numerical features, and most-frequent imputation followed by One-Hot Encoding for the categorical features.

The preprocessing steps were integrated into a full pipeline along with the regression models. This ensures that all the transformations were consistently applied during training and testing, reducing the risk of data leakage and improving reproducibility.

4.2 Model Training

We trained and evaluated multiple regression models to predict Global_Sales including Linear Regression, Decision Tree Regressor, and Random Forest Regressor. These models were selected to compare simple linear approaches against more flexible non-linear ones.

Linear Regression was used as a solid baseline model to capture linear relationships observed in the data between features and global sales. It provided a simple and interpretable benchmark for comparison.

The Decision Tree Regressor was trained to model non-linear relationships by recursively splitting the data based on feature values. We used the model with a max depth of 10 to reduce overfitting.

Lastly, the Random Forest Regressor was used to improve on performance and stability. It builds multiple decision trees and averages their predictions to reduce variance and improve generalization.

All models were trained using the same training dataset and preprocessing pipeline to ensure fair comparison.

4.3 Model Evaluation & Testing

Model performance was evaluated using Root Mean Squared Error (RMSE), because it measures the average magnitude of prediction error. It penalizes larger errors by squaring it, making it effective for assessing the accuracy of regression models involving sales data.

To obtain a reliable estimate of model performance, we used 5-fold cross-validation on the training set for each model. This approach splits the data into five subsets, trains the model on four subsets, and validates it on the remaining one, and repeats the process on the other folds. The average RMSE was used to compare model performance to reduce the impact of potential variability from a single train-test.

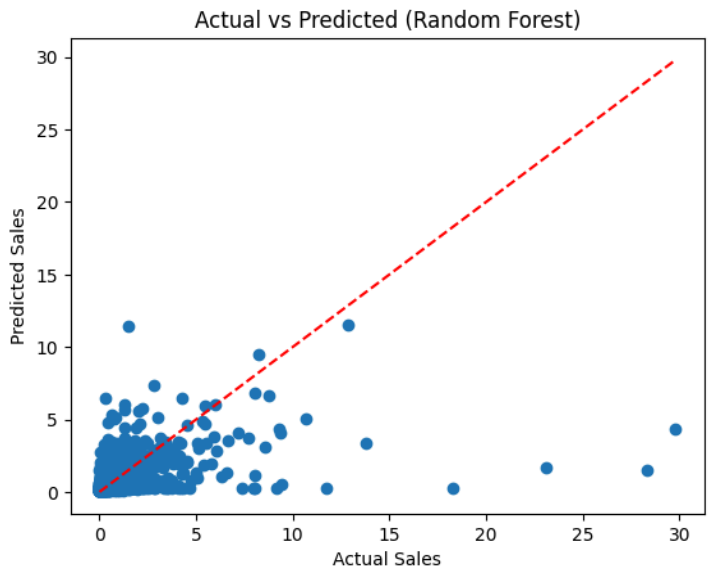
During evaluation, we observed that the Random Forest Regressor consistently produced more stable and lower RMSE values compared to the other models. The Decision Tree interestingly showed a higher variance than even the Linear Regression model, suggesting that the skewed distribution of *Global_Sales* affected its ability to predict trends. Linear Regression provided a good baseline but was less effective in capturing relationships in the data than the Random Forest Regressor. The lower RMSE of the Random Forest indicates that it produces more accurate predictions on average compared to the other two models, making it the most reliable for this task.

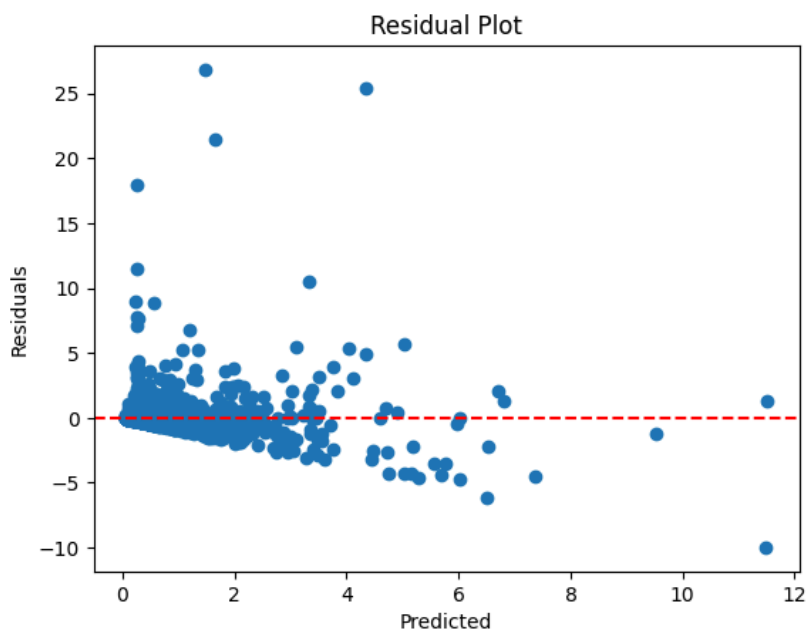
```
... Linear Regression AVG RMSE: 1.440758617570409
    Decision Tree AVG RMSE: 1.5569898498635641
    Random Forests AVG RMSE: 1.337350874327091
```

After identifying Random Forest as the best-performing model type, we continued to fine tune the model utilizing 5-fold cross-validation with grid search, lowering the RMSE by 15% to 1.1400. We found that from the given parameter combinations, max depth=10, min_samples_leaf=2, min_samples_split= 2, n_estimators=100 to be the best performing.

```
param_grid = {
    "regressor__n_estimators": [100, 200],
    "regressor__max_depth": [10, 15, None],
    "regressor__min_samples_split": [2, 5],
    "regressor__min_samples_leaf": [1, 2],
}
```

Evaluation Visuals:





The Actual vs Predicted plot shows that predictions generally follow the y=x line, however, larger deviations occur at higher sales values, suggesting that the model struggles in predicting extreme outliers, which is consistent with the skewed distribution of *Global_Sales*. The residual plot further supports this, with an increased spread for larger predicted values.

5. Results, Code Output & Interpretation

```
print(f"Linear Regression AVG RMSE: {lr_rmse_scores.mean()}")
print(f"Decision Tree AVG RMSE: {dt_rmse_scores.mean()}")
print(f"Random Forests AVG RMSE: {rf_rmse_scores.mean()}")

results = pd.DataFrame({
    "Model": ["Linear Regression", "Decision Tree", "Random Forest"],
    "Avg CV RMSE": [lr_rmse_scores.mean(), dt_rmse_scores.mean(), rf_rmse_scores.mean()]
})
```

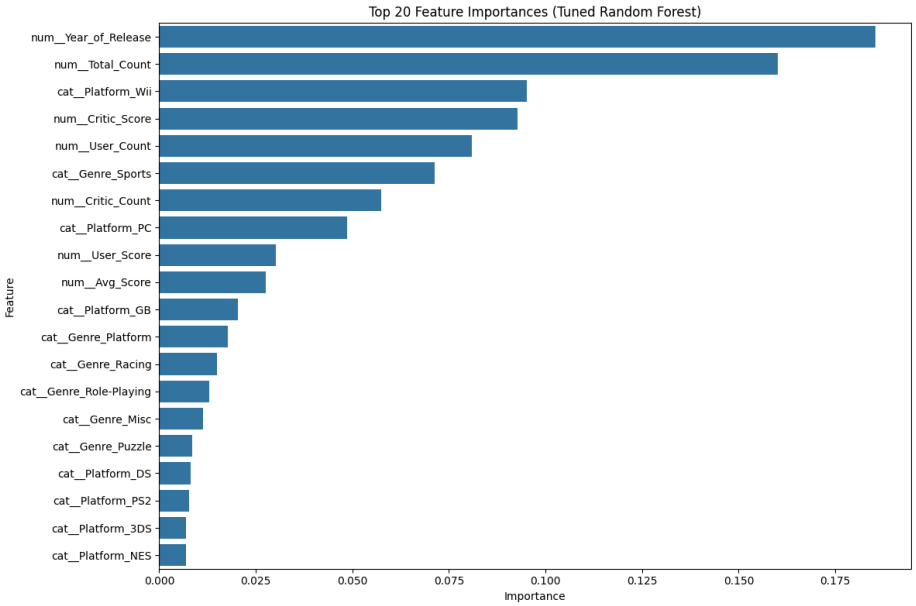
```
Linear Regression AVG RMSE: 1.440758617570409
Decision Tree AVG RMSE: 1.5569898498635641
Random Forests AVG RMSE: 1.337350874327091
```

```
y_pred = best_model.predict(X_test)
rft_rmse = np.sqrt(mean_squared_error(y_test, y_pred))
print("Tuned Test RMSE:", rft_rmse)
```

```
Tuned Test RMSE: 1.1382452091433128
```

The performance of the three regression models shows that the Random Forest model outperformed the others due to its lower average RMSE. This comparison shows that more flexible models are better suited for predicting global sales. The RMSE scores for these models are measured in millions of copies sold, meaning that after hyperparameter tuning the Random Forest model improved to an average prediction error of about 1.14 million units.

The Actual vs Predicted plot shows that most predictions follow the expected trend which shows that the model performs reasonably well overall. However larger errors appear for games with high sales, likely due to the highly skewed distribution of global sales.



Feature importance analysis revealed that variables such as Year_of_Release, and Total_Count, had the strongest influence on predictions, even more than variables that were expected to be more important like Critic_Score and User_Count.

These results directly answer our research questions by showing that while critic scores have some predictive power, they are not the strongest indicators of global sales. Factors such as release year, platform, and overall popularity play a larger role, confirming that commercial success is influenced by multiple variables beyond review scores.

6. Instructions for Running the Project

6.1 Interactive Plot

Requirements: Python Installed

Step 1: Install the zip file from the github repository

Step 2: Extract the file somewhere of your choice

Step 3: Go to the /Project Assignments/Q5 Interactive Plot/ folder

Step 4: Open a terminal in the folder by right clicking in the folder and clicking "Open in Terminal"

Step 5: Run "py -m pip install panel plotly pandas numpy" (without the quotation marks)

Step 6: Run "py -m panel serve app.py --show" (without the quotation marks)

6.2 Exploration Plots & Machine Learning Notebooks

Step 1: In Google Drive, create a folder called “CS133” in your “MyDrive” directory.

Step 2: Place the datafile csv in the “CS133” folder.

Step 3: Open the ExplorationPlots and MachineLearning ipynb files in Google Colab.

Step 4: Click “Run all” at the top left corner.

7. References

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